

MOHAMMAD MASRUR HOSSAIN

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PROFESSIONAL SUMMARY

Data-driven Semiconductor Metrology & Equipment Engineer with 5+ years of hands-on cleanroom experience, including nearly 4 years in industry specializing in advanced packaging workflows, die-level characterization, and hardware process optimization. Experienced in First-of-a-Kind (FOAK) equipment evaluation, vendor collaboration, and root cause analysis using SPC, DOE, FMEA, and FDC methodologies. Proven track record in sustaining tool health, mitigating defect issues, and seamlessly transferring stable processes from R&D/TD to High-Volume Manufacturing (HVM).

TECHNICAL SKILLS

- **Packaging Technology:** EMIB-T, Hybrid bonding.
- **Semiconductor & Packaging Equipment:** Metrology Toolsets, Grinding & Polishing Systems, Taper/De-taper Tools, Overlay & Warpage Systems, Tool Health & Recipe Optimization.
- **Process & Yield Engineering:** Defect Analysis (Chipping, Warpage, Cracking, Spacing), Failure Mode & Effects Analysis (FMEA), Root Cause Analysis, Statistical Process Control (SPC), Design of Experiments (DOE), Yield Enhancement.
- **Software & Data Engineering Tools:** JMP, Klarity, Python (NumPy, SciPy), ImageJ, ANSYS, SolidWorks, EBSILON®.
- **Materials Characterization:** XRD, SEM, TGA, FTIR, Raman Spectroscopy, BET Surface Area Analysis.

PROFESSIONAL EXPERIENCE

Intel Corporation — Metrology Equipment Development Engineer

May 2024 – Present

- Performed hands-on defect characterization and failure analysis on wafer-level inspection tools, developing Brightfield (BF) and Darkfield (DF) optical recipes to detect front/backside chipping, warpage, cracking, epoxy height, die-to-die bonding alignment, and spacing variations.
- Led Technology Development (TD) wafer-level metrology recipe creation and optimization across EMIB-T and Hybrid Bonding architectures, collaborating with Integration, Automation, and HVM teams to qualify robust modules for New Process Introductions (NPI).
- Coordinated continuous improvement projects (CIP) with OEM suppliers to optimize metrology hardware performance, increase tool availability, and reduce downtime in high-volume production environments.
- Designed and executed Design of Experiments (DOE) and Statistical Process Control (SPC) frameworks in JMP and Klarity to evaluate tool capability, monitor equipment health, and eliminate process excursions across advanced packaging lines
- Authored standard operating procedures (SOPs), BKM's, and equipment specifications to ensure recipe repeatability, tool-to-tool matching, and seamless module scaling across manufacturing sites.

Intel Corporation — Packaging R&D Sustaining Engineer

April 2023 – May 2024

- Oversaw tool availability, recipe stability, and consistent throughput across grinding, polishing, taper, and de-taper operations in advanced packaging environments.
- Utilized model-based problem solving and SPC trends to perform first-level hardware troubleshooting on tool alarms, reducing downtime and protecting die yield.
- Partnered with cross-functional equipment, integration, and process development teams to resolve tool excursions, data gaps, and recipe inconsistencies to maintain 24/7 operational continuity.

NRG Laboratory at the University of Washington — Research Assistant

Sept 2021 – March 2023

- Engineered a high-throughput, sub-second MOF synthesis platform combining solution- and vapor-phase delivery, establishing precise control over nucleation and growth kinetics.
- Characterized crystal structure, morphology, and thermal stability (XRD, SEM, TEM, TGA) to correlate equipment process conditions with material quality and structural integrity.
- Developed scalable synthesis frameworks across diverse MOF systems, ensuring robust process consistency and repeatability for advanced materials applications.

EDUCATION

MSc in Mechanical Engineering | University of Washington, Seattle | CGPA: 3.68

Sept 2021 – March 2023

BSc in Mechanical Engineering | Islamic University of Technology, Bangladesh | CGPA: 3.83

Jan 2017 – March 2021